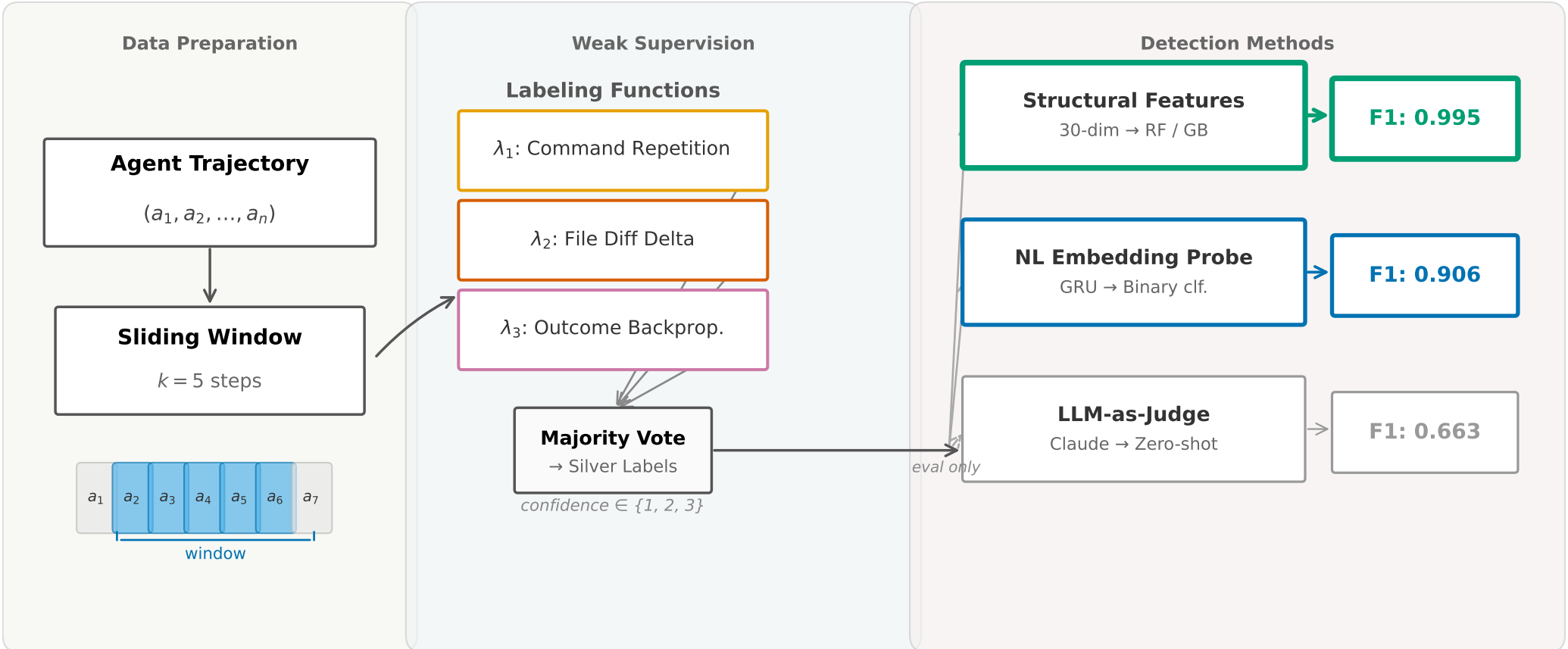
Data Preparation
Agent Trajectory
$(a_1, a_2, \ldots, a_n)$
Sliding Window
$k = 5$ steps
a_1 a_2 a_3 a_4 a_5 a_6 a_7
window
Weak Supervision
Labeling Functions
λ_1: Command Repetition
λ_2: File Diff Delta
λ_3: Outcome Backprop.
Majority Vote
→ Silver Labels
confidence ∈ {1, 2, 3}
Detection Methods
Structural Features
30-dim → RF / GB
F1: 0.995
NL Embedding Probe
GRU → Binary clf.
F1: 0.906
LLM-as-Judge
Claude → Zero-shot
F1: 0.663
eval only